Melika Baghi

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Summary

Ph.D. Candidate in Industrial & Systems Engineering at Georgia Tech with an M.S. in Statistics. I build machine learning and statistical models for high-dimensional, structured and incomplete data and ship them as reproducible, documented code, across healthcare analytics, scientific computing and sequential decision making. Seeking data science, machine learning and applied scientist roles.

Technical Skills

Programming & data: Python, SQL, R, PySpark, MATLAB, NumPy, pandas, SciPy

Machine learning: scikit-learn, XGBoost, PyTorch, deep learning, representation and multimodal learning, diffusion models

Statistics & modeling: statistical modeling, nonparametric statistics, Gaussian processes, design of experiments, uncertainty quantification, calibration, conformal prediction

Simulation & computing: mixed-integer programming, AnyLogic (agent-based + discrete-event), Simulink, Monte Carlo, Git, Linux, GPU/HPC, SLURM

Experience

Graduate Researcher — Georgia Institute of Technology *Aug 2023 – Present*

H. Milton Stewart School of Industrial & Systems Engineering · Advisors: K. Paynabar, X. Liu

- Develop ML and statistical methods for high-dimensional data — gradient boosting, deep learning, Gaussian processes, multimodal learning, uncertainty quantification — end to end from formulation through validation, reproducible code and publication.
- Built a constrained gradient-boosting estimator that enforces a manifold validity bound during fitting, cutting reconstruction error 5× against interpolation baselines; released as an installable open-source package.
- Built a tensor-structured surrogate for spatiotemporal simulation output, achieving 35.6×–564× complexity reduction and 4×–108× speedup while matching baseline accuracy within 0.35 percentage points.

Head Teaching Assistant — ISyE 6525: High-Dimensional Data Analytics *Jan 2024 – Present*

Georgia Institute of Technology

- Lead the TA team for a graduate course of 150+ students per semester on dimensionality reduction, sparse modeling and high-dimensional inference. Named **Outstanding Online TA**, 2025.

Course Instructor — Artificial Intelligence Laboratory *Aug 2022 – Aug 2023*

Royan Institute, Tehran

- Designed and taught a project-based applied AI curriculum in clinical healthcare contexts.

Selected Projects

- **Hospital capacity and patient flow** — hybrid agent-based and discrete-event simulation of a multi-specialty hospital serving medical tourists and local patients. Compatible-section bed sharing cut the mean admission queue **43%** (25.9 to 14.8 days, 30 replications, $p < 0.001$). Screened 16 operational factors with a 256-run fractional factorial design. *AnyLogic, Python*
- **Adaptive multimodal prediction under missing data** — certifies a prediction when the evidence already collected determines the decision, and otherwise abstains and recommends which modality to acquire next, with finite-sample coverage guarantees. Evaluated on multimodal benchmarks. *Python, PyTorch*
- **Clinical risk and drug safety from EHR** — adverse drug event detection from longitudinal EHR using temporally-aware patient embeddings over medication trajectories, labs and comorbidities; compared regularized regression, tree ensembles and neural models with calibration and fairness audits across demographic strata. Related work: contrastive multimodal phenotype modeling over transcriptomics, EHR and imaging. *Python*
- **Predictive maintenance and anomaly detection** — end-to-end pipeline for sensor-based fault detection: ARIMA and LSTM forecasting, ensemble classifiers over multivariate telemetry, and hybrid Weibull–state-space degradation modeling with online retraining, giving over **15 hours** of advance fault warning; deployed on Spark for low-latency inference. *Python, PySpark*
- **Learning from delayed feedback** — bandit algorithm that pools one delayed outcome across every action that could have produced it, at a cost set by an effective dimension measuring action overlap. *Python*

Education

Ph.D., Industrial & Systems Engineering (System Informatics & Control), Georgia Institute of Technology *Aug 2023 – Present*

M.S., Statistics, Georgia Institute of Technology *Aug 2023 – May 2026*

M.S., Industrial Engineering — Healthcare Systems, Amirkabir University of Technology; GPA 4.0/4.0 *2021 – 2023*

B.S., Industrial Engineering (GPA 3.91/4.0) and **B.S., Aerospace Engineering** (GPA 3.79/4.0), Amirkabir University of Technology *2016 – 2021*

Selected Publications

Four preprints and manuscripts under review in machine learning and healthcare operations, including arXiv:2605.04130 (constrained gradient boosting), arXiv:2608.15520 (adaptive modality acquisition), arXiv:2605.19139 (hospital simulation) and arXiv:2609.01761 (delayed bandits). Full list: Google Scholar.